

CSPM Auto Remediation using AWS

Name: Sonali Gupta

1. Objective

The objective of this project is to implement a Cloud Security Posture Management (CSPM) solution that automatically detects and remediates insecure configurations in Amazon S3 buckets. The system ensures that newly created buckets are secured and that the user is notified through email alerts.

2. Architecture Overview

The solution is implemented using the following AWS services:

- AWS CloudTrail – Captures API activity
- Amazon EventBridge – Detects S3 bucket creation events
- AWS Lambda – Performs automated remediation
- Amazon SNS – Sends email notifications
- Amazon S3 – Resource being monitored

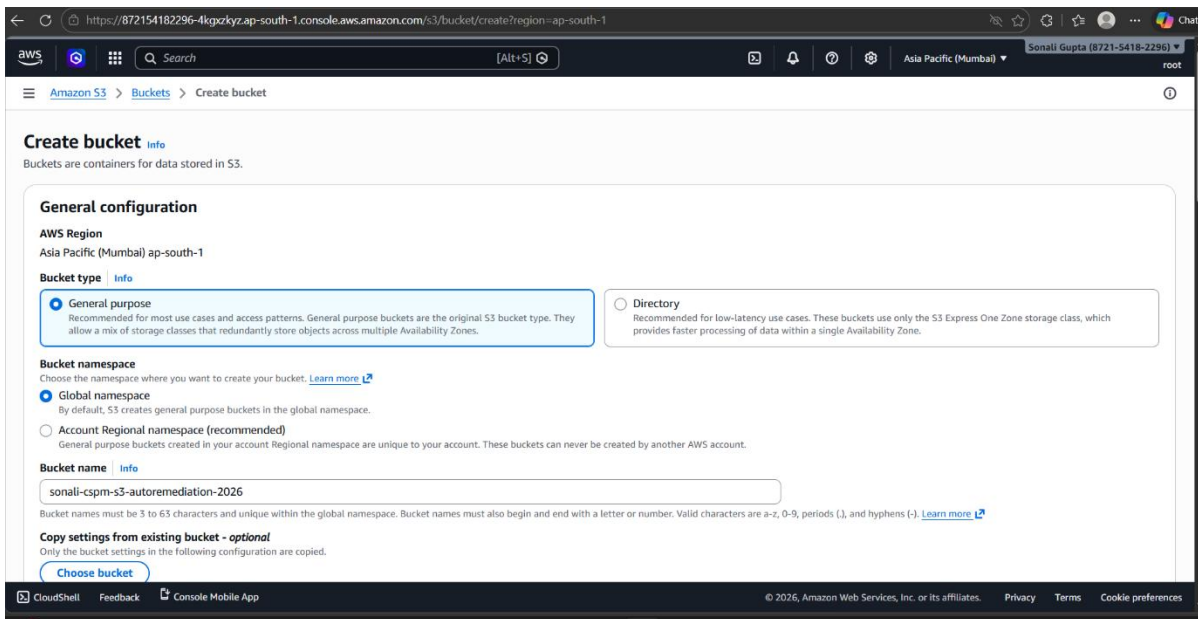
3. Workflow

1. A new S3 bucket is created.
2. CloudTrail logs the API call.
3. EventBridge detects the CreateBucket event.
4. EventBridge triggers the Lambda function.
5. Lambda function:
 - Extracts the bucket name
 - Applies security configuration (blocks public access)
 - Sends notification via SNS
6. Email alert is delivered to the user.

4. Implementation

4.1 S3 Bucket Creation

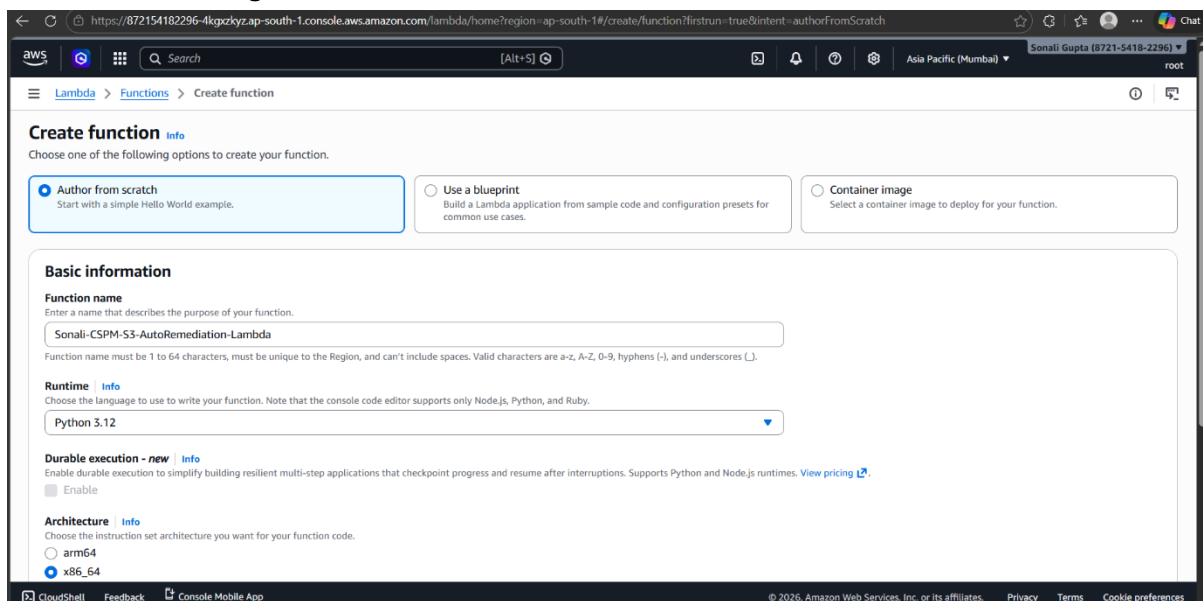
Test buckets were created to simulate real-world scenarios and validate the system behavior.

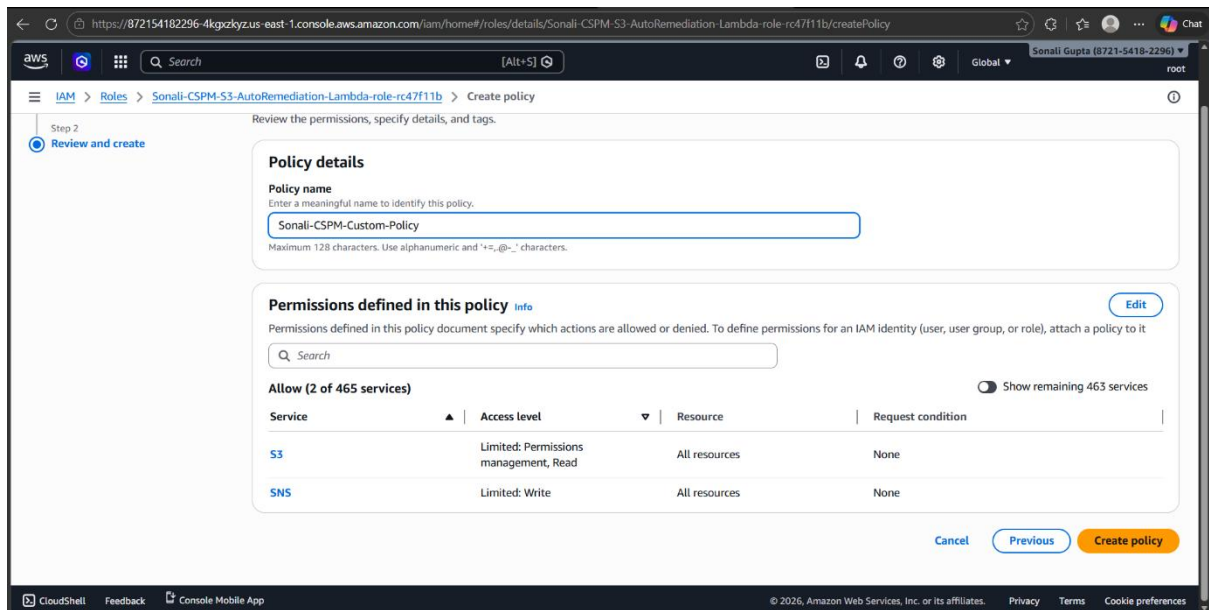


4.2 Lambda Function

A Lambda function was developed using Python to automate remediation. The function:

- Extracts the bucket name from the event
- Applies security settings to block public access
- Sends an alert using SNS



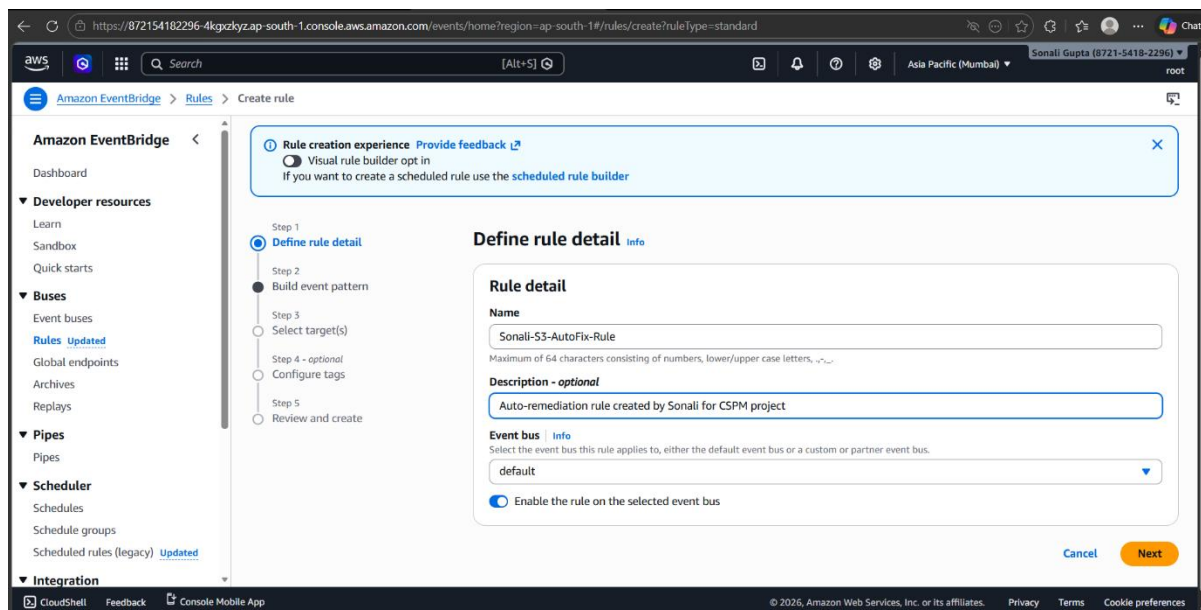


4.3 EventBridge Rule

An EventBridge rule was configured to detect S3 bucket creation events.

Event Pattern:

```
{
  "source": ["aws.s3"],
  "detail-type": ["AWS API Call via CloudTrail"],
  "detail": {
    "eventSource": ["s3.amazonaws.com"],
    "eventName": ["CreateBucket"]
  }
}
```



4.4 CloudTrail Configuration

CloudTrail was enabled with a multi-region trail to capture management events, specifically S3 API calls required for event detection.

aws

Asia Pacific (M

Sonali Gupta (8721-5418...

root

CloudTrail > Trails > Create trail

Step 1

☒ Choose trail attributes

Step 2

☐ Choose log events

Step 3

☐ Review and create

Choose trail attributes

General details

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name

Enter a display name for your trail.

sonali-trail

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization

To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location [Info](#)

☒ Create new S3 bucket
Create a bucket to store logs for the trail.

☐ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket and folder

Enter a new S3 bucket name and folder (prefix) to store your logs.

CloudShell

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Trail successfully created

Trails

Copy events to Lake  Delete **Create trail**

	Name ▲	Home region ▼	Multi-region trail ▼	ARN ▼	Insights ▼	Organization trail ▼	S3 bucket ▼	Log file prefix ▼	CloudWatch Logs log group ▼	Status ▼
☐	sonali-trail	Asia Pacific (Mumbai)	Yes	arn:aws:cloudtrail:ap-south-1:872154182296:trail/sonali-trail	Disabled	No	aws-cloudtrail-logs-872154182296-776e0c4c12	-	-	 Logging

4.5 SNS Notification

An SNS topic was created and configured with an email subscription. The subscription was confirmed to receive alerts generated by the Lambda function.

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

Search

[Alt+S]

Asia Pacific (Mumbai)

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root

Amazon SNS

Topics

Sonali-CSPM-Alerts-Topic

Topic Sonali-CSPM-Alerts-Topic created successfully.

You can create subscriptions and send messages to them from this topic.

Notifications

0

0

1

1

0

0

Sonali-CSPM-Alerts-Topic

Edit

Delete

Publish message

Details

Name

Sonali-CSPM-Alerts-Topic

ARN

arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-Topic

Display name

-

Type

Standard

Topic owner

872154182296

Subscriptions

Access policy

Data protection policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption

Tags

Subscriptions (0)

Edit

Delete

Request confirmation

Confirm subscription

Create subscription

Search

1

ID

Endpoint

Status

Protocol

No subscriptions found.

CloudShell

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Amazon SNS

Subscriptions

Create subscription

New Feature

Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create subscription

Details

Topic ARN

arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-Topic

Protocol

The type of endpoint to subscribe

Email

Endpoint

An email address that can receive notifications from Amazon SNS.

sonalig4021997@gmail.com

After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional

[Info](#)

This policy filters the messages that a subscriber receives.

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https://872154182296-4kgzkyz.ap-south-1.console.aws.amazon.com/sns/v3/home?region=ap-south-1#/subscription/arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-Topic...

🔍 Search

[Alt+S]

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Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

🔔 Subscription to Sonali-CSPM-Alerts-Topic created successfully.
The ARN of the subscription is arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-Topic:8225d485-dbf8-4cf0-b0eb-d0db0b7016a1.

Notifications 📶 0 ⚠️ 0 🟢 1 🟡 1 📵 0

Subscription: 8225d485-dbf8-4cf0-b0eb-d0db0b7016a1

Edit Delete

Details

ARN

arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-Topic:8225d485-dbf8-4cf0-b0eb-d0db0b7016a1

Endpoint

sonalig4021997@gmail.com

Topic

Sonali-CSPM-Alerts-Topic

Subscription Principal

arn:aws:iam::872154182296:root

Status

🕒 Pending confirmation

Protocol

EMAIL

Subscription filter policy

Redrive policy (dead-letter queue)

Subscription filter policy

Info

This policy filters the messages that a subscriber receives.

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10:27 AM



Subscript...
com



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-
Topic:8225d485-dbf8-4cf0-b0eb-d0db0b7016a1

If it was not your intention to subscribe, [click here to unsubscribe](#).



4.6 IAM Role

An IAM role was assigned to the Lambda function with permissions to:

- Modify S3 bucket configurations
- Publish messages to SNS

Step 2
[Review and create](#)

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.

Maximum 128 characters. Use alphanumeric and '+=, @, _' characters.

Permissions defined in this policy [Info](#) [Edit](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Allow (2 of 465 services) [Show remaining 463 services](#)

Service	Access level	Resource	Request condition
S3	Limited: Permissions management, Read	All resources	None
SNS	Limited: Write	All resources	None

[Cancel](#) [Previous](#) [Create policy](#)

5. Testing and Output

The system was tested by creating a new S3 bucket. Upon creation:

- The event was captured by CloudTrail
- EventBridge triggered the Lambda function
- The Lambda function applied security settings
- An email notification was successfully received

Sample Output:

- Bucket: final-test-bucket-789
- Action: Public access blocked
- Notification: Email alert received

CloudWatch

Log management

/aws/lambda/Sonali-CSPM-S3-AutoRemediation-Lambda

2026/03/29/[LATEST]08d45ac3959147b99a17ffb32f28701

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

Metrics

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events.

Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone

Display

Timestamp	Message
No older events at this moment. Retry	
2026-03-29T06:57:54.677Z	INIT_START Runtime Version: python:3.12.mainlinev2.v4 Runtime Version ARN: arn:aws:lambda:ap-south-1::runtime:ffd8d8a0ee4551835e06b2...
2026-03-29T06:57:55.172Z	START RequestId: 0cfa36cd-22f0-4cf9-b6d1-1684f6196a24 Version: \$LATEST
2026-03-29T06:57:55.173Z	Event received: {'version': '0', 'id': 'bd045f38-b1f9-b5e3-ff0c-42370fd22d26', 'detail-type': 'AWS API Call via CloudTrail', 'source...
2026-03-29T06:57:55.173Z	Processing bucket: final-test-bucket-789
2026-03-29T06:57:55.999Z	END RequestId: 0cfa36cd-22f0-4cf9-b6d1-1684f6196a24
2026-03-29T06:57:55.999Z	REPORT RequestId: 0cfa36cd-22f0-4cf9-b6d1-1684f6196a24 Duration: 826.21 ms Billed Duration: 1318 ms Memory Size: 128 MB Max Memory U...
No newer events at this moment. Auto retry paused. Resume	

CloudShell

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12:29 PM

VoWiFi 109 K/S 82



Sonali CSPM Alert

Inbox



AWS No... 12:27 PM

to me ▾



CSPM ALERT

Student: Sonali

Project: CSPM Auto Remediation

Bucket: final-test-bucket-789

Region: ap-south-1

Issue: PUBLIC ACCESS DETECTED

Fix: Converted to PRIVATE

Trigger: AWS Lambda

--

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:

[https://sns.ap-south-1.amazonaws.com/unsubscribe.html?](https://sns.ap-south-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-Topic:8225d485-dbf8-4cf0-b0eb-d0db0b7016a1&)

SubscriptionArn=arn:aws:sns:ap-south-1:872154182296:Sonali-CSPM-Alerts-Topic:8225d485-dbf8-4cf0-b0eb-d0db0b7016a1&

↩ Reply

➦ Forward



6. Conclusion

The project successfully demonstrates an automated CSPM solution that ensures S3 bucket security at the time of creation. The integration of AWS services enables real-time detection, automated remediation, and user notification.

7. Resource Cleanup

After completing the implementation and testing, all resources were deleted to avoid unnecessary AWS charges. These include:

- S3 buckets
- Lambda function
- EventBridge rule
- SNS topic
- CloudTrail trail
- IAM roles
- KMS keys

aws [Alt+S] Search Asia Pacific (Mumbai) Sonali Gupta (8721-5418-2296) root

Amazon EventBridge > Rules

1 rule(s) deleted successfully

Rules

A rule watches for specific types of events. When a matching event occurs, the event is routed to the targets associated with the rule. A rule can be associated with one or more targets.

Select event bus

Event bus
Select or enter event bus name
default

Event pattern rules [Delete] [Enable] [Edit] [CloudFormation Template] [Create rule]

We have moved Scheduled rules in your account to 'Scheduler' section, where you can view, create and edit a scheduled rule. [Go to Scheduled rules]

Find rules [Any status]

Name	Status	Type	Event bus	ARN	Description
------	--------	------	-----------	-----	-------------

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IAM > Roles

Roles (2/5)

An IAM role is an identity...

Delete 2 roles?

Delete 2 roles permanently? This will also delete all their inline policies and any attached instance profiles.

Role name	Last activity
Sonali-CSPM-53-AutoRemediation-Lambda-role-rc4...	12 minutes ago
Amazon_EventBridge_Invoke_Lambda_2141750978	23 minutes ago

Note: Recent activity usually appears within 4 hours. Data is stored for a maximum of 365 days, depending when your region began supporting this feature. [Learn more]

This action cannot be undone.

To confirm deletion, enter delete in the text input field.
delete

[Cancel] [Delete]

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KMS > Customer managed keys

Successfully scheduled deletion for key `arn:aws:kms:ap-south-1:872154182296:key/33a305f2-2740-40c4-aab9-0b54a5808c44`

Notifications [0] [0] [2] [0] [0] [0]

Customer managed keys (1)

Filter keys by properties or tags

Aliases	Key ID	Status	Key type	Key spec	Key usage
alias/sonali-cspm-clo...	33a305f2-2740-40c4...	Pending deletion	Symmetric	SYMMETRIC_DEFAULT	Encrypt and decrypt

aws

Search

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KMS

Customer managed keys

Schedule key deletion

Key Management Service (KMS)

AWS managed keys

Customer managed keys

Custom key stores

AWS CloudHSM key stores

External key stores

Deleting a key makes all data encrypted under that key unrecoverable. [Learn more](#)

You can create an Amazon CloudWatch alarm that alerts you about any attempts to use the key(s) during the waiting period. [Learn more](#)

Waiting period

You can cancel deletion any time before the waiting period ends. After the waiting period ends, AWS KMS deletes the key(s). [Learn more](#)

Waiting period (in days)

7

Enter a waiting period between 7 and 30 days.

Keys to delete

The following keys will be scheduled for deletion.

Aliases	Key ID	Key usage	Regionality
alias/sonali-cspm-cloudtrail	33a305f2-2740-40c4-aab9-0b54a5808c44	Encrypt and decrypt	Single Region

Confirmation

☒ Confirm that you want to schedule these keys for deletion after a 7 day waiting period.

Cancel

Schedule deletion

aws

Search

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KMS

Customer managed keys

Key Management Service (KMS)

AWS managed keys

Customer managed keys

Custom key stores

AWS CloudHSM key stores

External key stores

Customer managed keys (1/1)

Filter keys by properties

☒ Aliases

☒ alias/sonali-cspm-cloudtrail

Disable key

Disabling a key prevents you from using it in any cryptographic operations. Also, disabling might prevent you from accessing resources in AWS services that use this key.

To monitor attempts to use the key while it is disabled, create an Amazon CloudWatch alarm. [Learn more](#)

To make the key usable again, enable the key.

Alias

Key ID

alias/sonali-cspm-cloudtrail

33a305f2-2740-40c4-aab9-0b54...

☒ Confirm that you want to disable this key.

Cancel

Disable key

Key actions

Create key

1

Key spec

Key usage

SYMMETRIC_DEFAULT

Encrypt and decrypt

aws

Search

[Alt+S]

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Amazon SNS

Subscriptions

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

Push notifications

Text messaging (SMS)

Subscription deleted successfully.

Notifications 0 0 1 1 0 0

Subscriptions (0)

Edit

Delete

Request confirmation

Confirm subscription

Create subscription

Search

1

ID

Endpoint

Status

Protocol

Topic

No subscriptions found

Create subscription

Amazon SNS

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Text messaging (SMS)

Topic Sonali-CSPM-Alerts-Topic deleted successfully.

Notifications 0 0 0 1 1 0

Topics (0)

Search

Edit Delete Publish message Create topic

< 1 >

Name	Type	ARN
No topics		
To get started, create a topic.		
Create topic		

https://872154182296-4kgzkyz.ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions

Search [Alt+S]

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root

Lambda

Functions

Successfully updated the function Sonali-CSPM-S3-AutoRemediation-Lambda.

Successfully deleted function: Sonali-CSPM-S3-AutoRemediation-Lambda

Functions (0)

Last fetched 29/3/2026, 12:46:41 pm

Actions

Create function

Search by attributes or search by keyword

< 1 >

Function name	Description	Package type	Runtime	Type	Last modified
There is no data to display.					

CloudTrail

Trails

Trail successfully deleted

Copy events to Lake Delete Create trail

Trails

< 1 >

Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group	Status
No trails									
No trails to display.									
Create trail									

CloudShell

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